



FREEDOM INTERNATIONAL SCHOOL

CERTIFICATE

This is to certify that

PARIDUL SUDARSHAN

has satisfactorily completed the activities and projects in
MATHEMATICS

prescribed by the Central Board of Secondary Education
at Freedom International School in the

Year 2020 - 2021

Signature of the teacher in charge of the batch

Date

Principal's Signature and Seal

Signature : 
Date : 11.02.2021

PRINCIPAL
FREEDOM INTERNATIONAL SCHOOL
BANGALORE

Name of the Candidate : PARIDUL SUDARSHAN

Roll Number : 18

Signature of the Internal Examiner : 

Signature of the External Examiner : _____

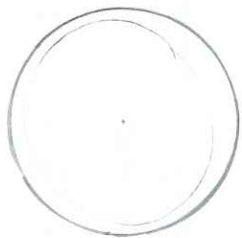
External Examiner Code : _____

Date of Practical Examination : 11.02.2021

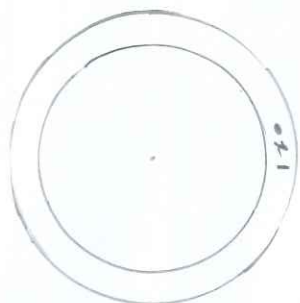




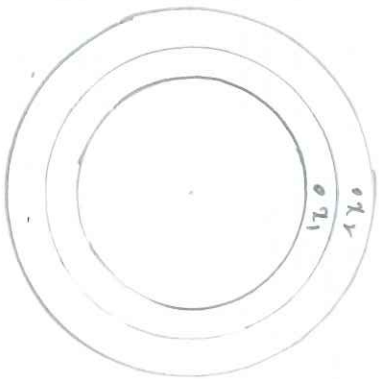
EXP 1:



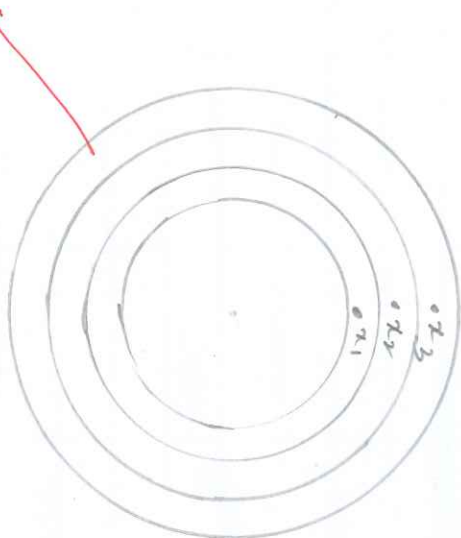
X_0



X_1



X_2



X_3

NUMBER OF SUBSETS OF A GIVEN SETObjective:

To find the number of subsets of a given set X , if it has n elements

Materials required:

white sheet, white pen, geometry box

Procedure:

Take X_0 an empty set. The number of possible subsets of X_0 is X_0 which is denoted by ϕ

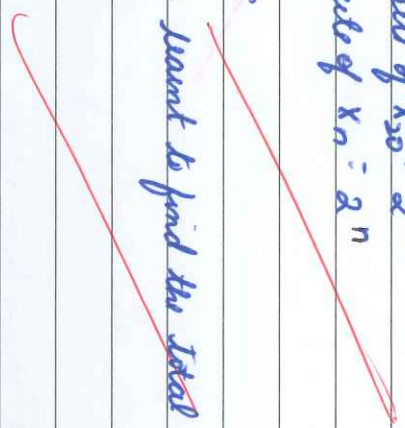
$$X_0 \Rightarrow 2^0 = 1$$

Take X_1 , one ^{with} element say x_1 . No. of subsets are $\phi, \{x_1\}$
No. of subsets: $X_1 \Rightarrow 2^1 = 2$

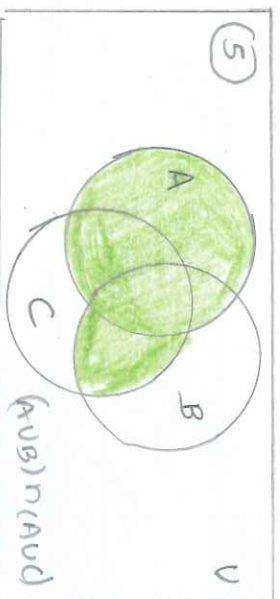
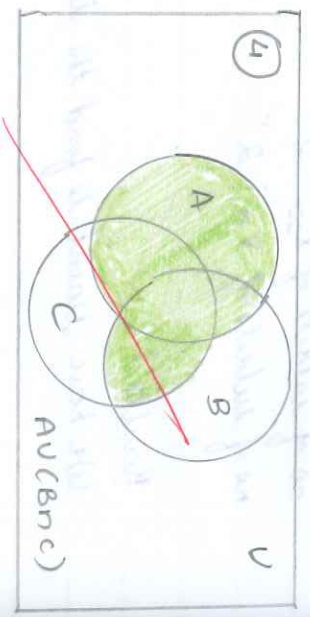
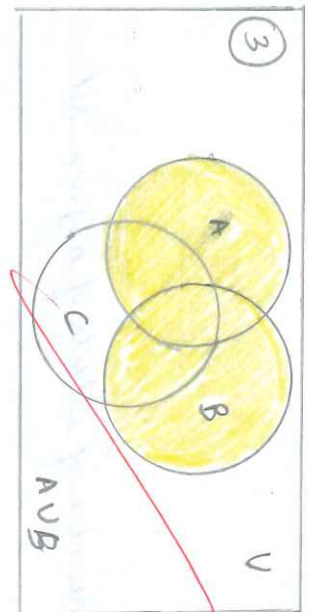
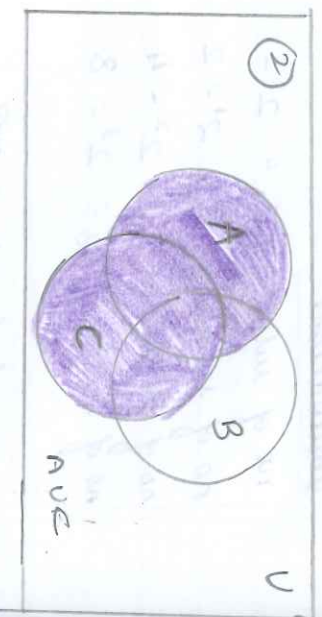
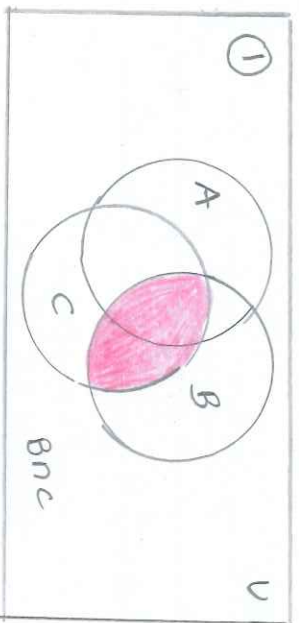
Take X_2 with 2 elements say x_1, x_2 . Subsets are $\phi, \{x_1\}, \{x_2\}, \{x_1, x_2\}$
No. of subsets: $X_2 \Rightarrow 2^2 = 4$

Take X_3 with 3 elements x_1, x_2, x_3 . Subsets are $\phi, \{x_1\}, \{x_2\}, \{x_3\}, \{x_1, x_2\}, \{x_2, x_3\}, \{x_1, x_3\}, \{x_1, x_2, x_3\}$
No. of subsets: $X_3 = 2^3 = 8$

Continuing like this we have number of subsets of set Z containing n elements $x_1, x_2, \dots, x_n$ is 2^n

Observations:no. of subsets of $X_0 = 2^0 = 1$ no. of subsets of $X_1 = 2^1 = 2$ no. of subsets of $X_2 = 2^2 = 4$ no. of subsets of $X_3 = 2^3 = 8$ ~~no. of subsets of $X_{20} = 2^{20}$
no. of subsets of $X_n = 2^n$~~ Results:~~We have learnt to find the total number of subsets of a given set.~~
Dr. H. S. D. S.

EXP 2:



DISTRIBUTIVE LAW FOR THREE GIVEN NON-EMPTY SETS

Objective:

Verify the distributive law for three given non empty sets A, B, C

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Materials required:

Hardboard, glazing box, thick sheet of paper, coloured pencil, pair of scissors, adhesive

Procedure:

Cut 5 rectangular sheets from paper. Paste them on the hardboard with the symbol U in the right top corner of each rectangle

Draw three circles and mark them as A, B, C in each of the 5 rectangle

Shade the portions as shown

Observations

Shaded portion in Fig 1 BnC

Shaded portion in Fig 2 AUC

Shaded portion in Fig 3 AUB

Shaded portion in Fig 4 AUCBnC

Shaded portion in Fig 5 (AUB) n (AUC)

Common shaded portion in 4 & 5 is AUCBnC

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

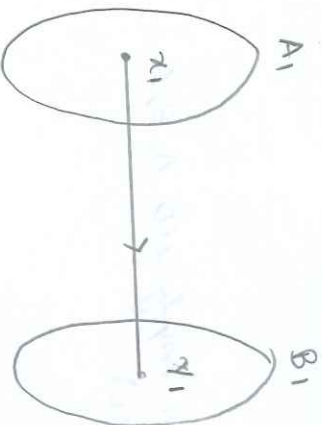
Hence the distributive law is verified

Result:

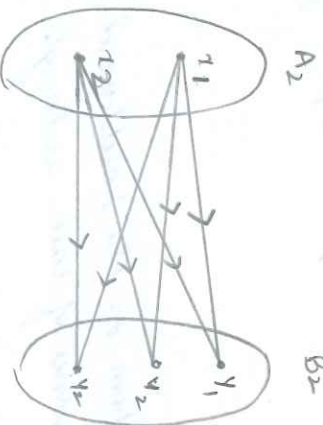
Distributive law for three given non empty sets is verified

Signature

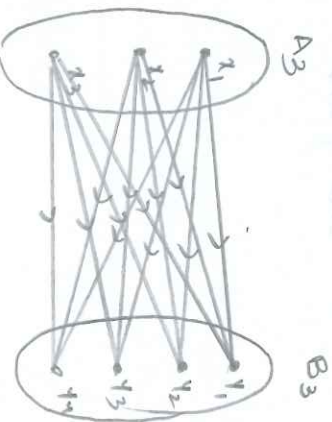
EXP 3 :



RELATION FROM
 A_1 TO B_1



RELATION FROM
 A_2 TO B_2



RELATION FROM
 A_3 TO B_3

NUMBER OF RELATIONS

Objective:

To verify that for two non empty finite sets A and B, "if $n(A) = m$ and $n(B) = n$, then total no. of relations from A to B is 2^{mn} ". That is $2^{n(A \times B)} = 2^{mn}$

Materials required:

white sheet, colour pencil - geometry box

Procedure:

Take set A, with element x_1 , set B, with element y_1 ,

Take set A, with x_1, x_2 and B, with y_1, y_2, y_3

Take set A, with x_1, x_2, x_3 and B, with y_1, y_2, y_3, y_4

Observation:

No. of elements, i.e. no. of elements in $A \times B$ of A and B is 1×1 and number of relations is 2^1

Similarly for A, and B $\Rightarrow 2 \times 3 \Rightarrow 2^6$

for A, and B $\Rightarrow 3 \times 4 \Rightarrow 2^{12}$

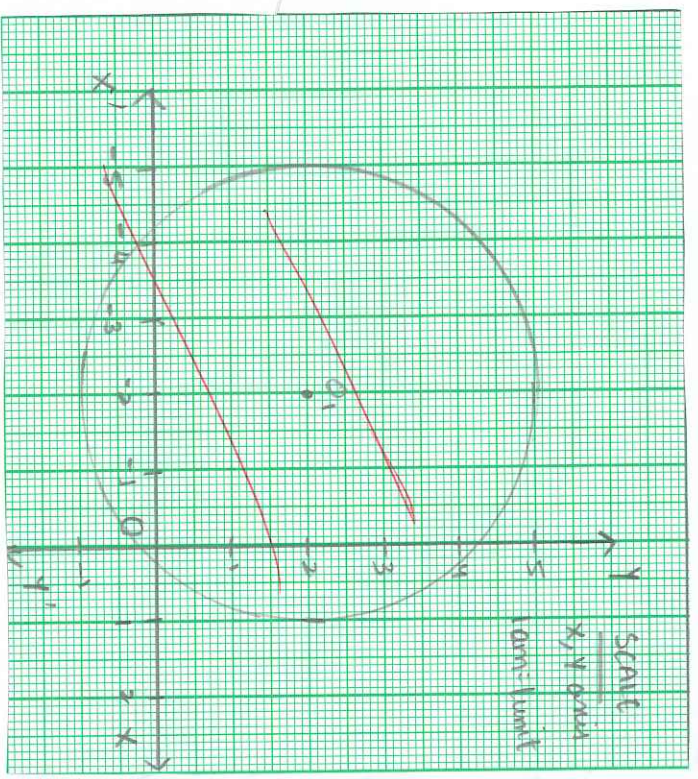
Hence, in the case of set A with m elements, set B with n elements, no. of relations is 2^{mn}

Result:

We have verified that for two non empty finite sets A and B, $n(A \times B) = mn$ and total no. of relations is 2^{mn}

Dr. Jyoti D. Patil

EXP 4:



Q. A circle is drawn with its center at the origin (0,0) and a radius of 5 units. A line segment is drawn from the point (-5, 0) on the x-axis to the point (0, 3) on the y-axis. Find the length of the line segment.

Solution: The length of the line segment is the distance between the points (-5, 0) and (0, 3). Using the distance formula, we get:

$$= \sqrt{(-5 - 0)^2 + (0 - 3)^2} = \sqrt{25 + 9} = \sqrt{34}$$

CIRCUMFERENCE AND AREA OF CIRCLE WHOSE EQUATION IS GIVEN

Objective:

To find the circumference and area of the circle whose equation is $x^2 + y^2 + 4x - 4y - 1 = 0$

Materials required:

sheet, compass, scale, calculator, pen

Procedure:

given equation: $x^2 + y^2 + 4x - 4y - 1 = 0$ ①
 standard equation: $x^2 + y^2 + 2gx + 2fy + c = 0$ ②

comparing ① & ②

$g = 2, f = -2, c = -1$

Centre = $(-g, -f) = (-2, 2)$

radius = $\sqrt{g^2 + f^2 - c} = \sqrt{9} = 3 \text{ units}$

Steps of construction:

Take a graph paper, mark the origin, mark the centre as $(-2, 2)$

Take $r = 3 \text{ cm}$, arc and draw circle

determine circumference and area

Observation table:

S.NO	Particulars	Observations
1	Equation: $x^2 + y^2 + 4x - 4y - 1 = 0$	
2	centre	$-2, 2$
3	radius	3 unit
4	circumference	19 unit
5	area	28.75 unit^2

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EXPERIMENT RESULT

Results:

for the given equation, centre $O(-3,2)$ and radius $R(3\text{cm})$ are determined
circumference = 19cm / unit

Area = 28.75 unit^2

Dr. H. S. S.

MEAN AND STANDARD DEVIATIONObjective:

To find mean and standard deviation of

x_i	10	15	18	20	25
f_i	3	2	5	3	2

Materials required:

calculator, practical workbook

Procedure:Mean:

Mean is the sum of all observations divided by number of observations

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum f_i}$$

Standard deviation:

It is the root mean square deviation when $A = \bar{x}$ so standard deviation is defined as the positive square root of the mean of the square of the deviation from arithmetic mean and is denoted by σ

$$\sigma = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{N}}$$

where $f_1 + f_2 + \dots + f_n = N$ Result:

Table for calculation of mean and standard deviation:

MEAN AND STANDARD DEVIATION

Objective :

To find mean and standard deviation of

x_i	10	15	18	20	25
f_i	3	2	5	8	2

Materials required :

calculator, practical workbook

Procedure :

Mean :

Mean is the sum of all observations divided by number of observations

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum f_i}$$

Standard deviation :

It is the root mean square deviation when $A = \bar{x}$ so standard deviation is defined as the positive square root of the mean of the square of the deviation from arithmetic mean and is denoted by σ

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where $f_1 + f_2 + \dots + f_n = N$

Result :

Table for calculation of mean and standard deviation:

S.No	Observed data	f_i	$f_i x_i$	$f_i (x_i - \bar{x})^2$
1	10	3	30	192
2	15	2	30	18
3	18	5	90	0
4	20	8	160	32
5	25	2	50	98

~~$$\text{mean} = \frac{\sum x_i f_i}{\sum f_i}$$~~

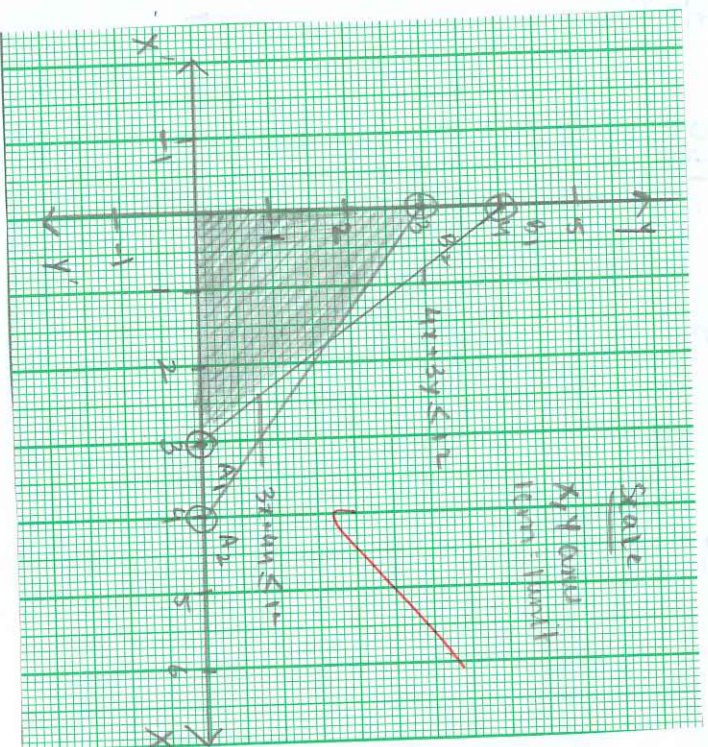
~~$$\bar{x} = \frac{\sum x_i f_i}{\sum f_i} = \frac{360}{20} = 18$$~~

~~$$\sum f_i (x_i - \bar{x})^2 = 340$$~~

~~$$s^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N} = \frac{340}{20} = 17$$~~

~~$$s = \sqrt{17} = 4.12$$~~

Divyanshu



Exp 6;

Scale
X: 1 cm
Y: 1 cm

SOLUTION SET OF LINEAR EQUATIONSObjective:

Exhibit graphically the solution set of the linear equations

$$3x + 4y \leq 12$$

$$4x + 3y \leq 12$$

$$x \geq 0$$

$$y \geq 0$$

Materials required:

graph paper, compass box, scale, calculator

Procedure:

convert inequalities to equations

$$3x + 4y = 12$$

$$4x + 3y = 12$$

$$x = 0$$

$$y = 0$$

Region represented by $3x + 4y \leq 12$, line $3x + 4y = 12$ meets coordinate axes at A (4, 0) B (0, 3). Join A and B. (0, 0) satisfies the equation so portion containing origin satisfies the equation

Region represented by $4x + 3y \leq 12$, line $4x + 3y = 12$ meets axes at A (3, 0) B (0, 4). Join A and B. (0, 0) satisfies the equation so portion containing origin satisfies the equation

Region represented by $x \geq 0$, $y \geq 0$ - first quadrant
Hence the shaded region represents the solution for the given set of linear inequalities

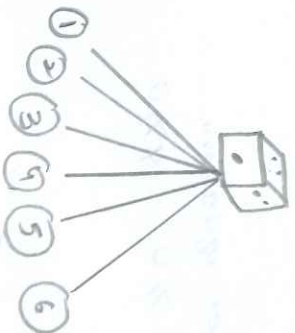
Result:

shaded region represents solution for given set of linear inequalities

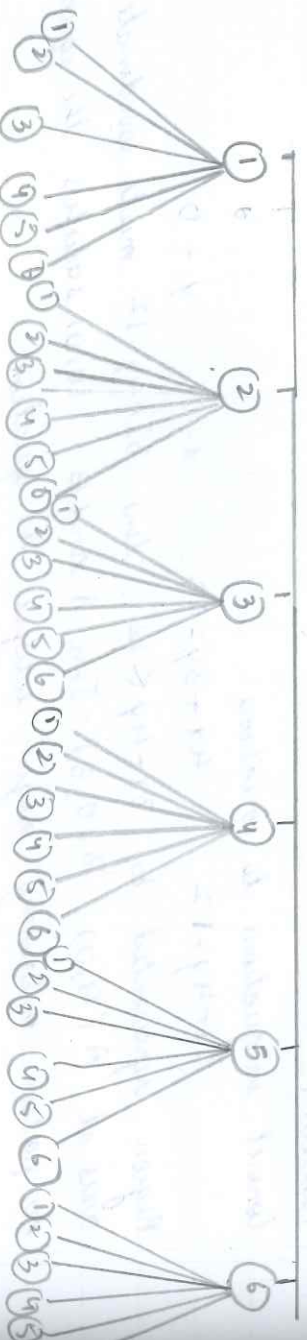
P. Murugesan

EXP 7:

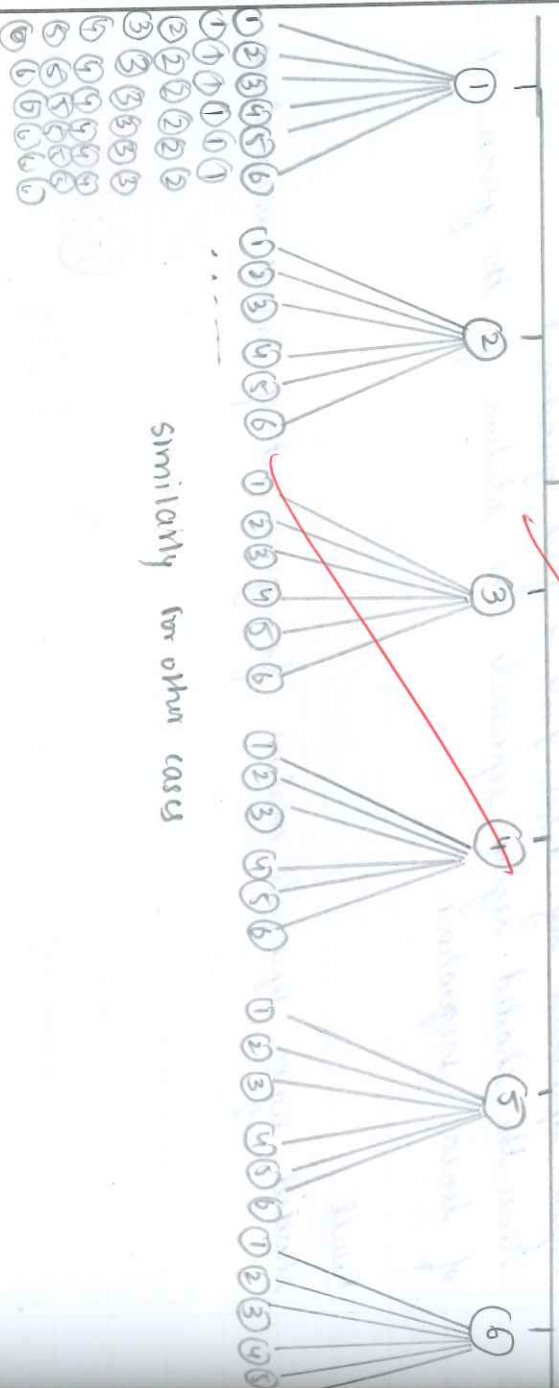
CASE 4:



CASE 2:



CASE 3:



similarly for other cases

SAMPLE SPACE

Objective : to generate a formula for the sample space when a die is rolled once, twice, three and so on.

Materials required :
die, sheets, pencil, plastic dice, worksheet

Pseudocode :
Throw a dice once. outcome will be 1, 2, 3, 4, 5 or 6
Make a tree diagram and write the sample space

Throw a die twice. Its outcome will be any of the 36 ways in the tree diagram

Repeat the experiment by throwing the die 3 times, write the tree diagram

Observations :

No. of elements in sample space when a die is thrown
once : 6^1 twice : 6^2 three : 6^3 four times : 6^4

Result :
we have found the formula for the sample space when a die is thrown a given no. of times

Signature

Exp 8:

For all graphs

Scale:

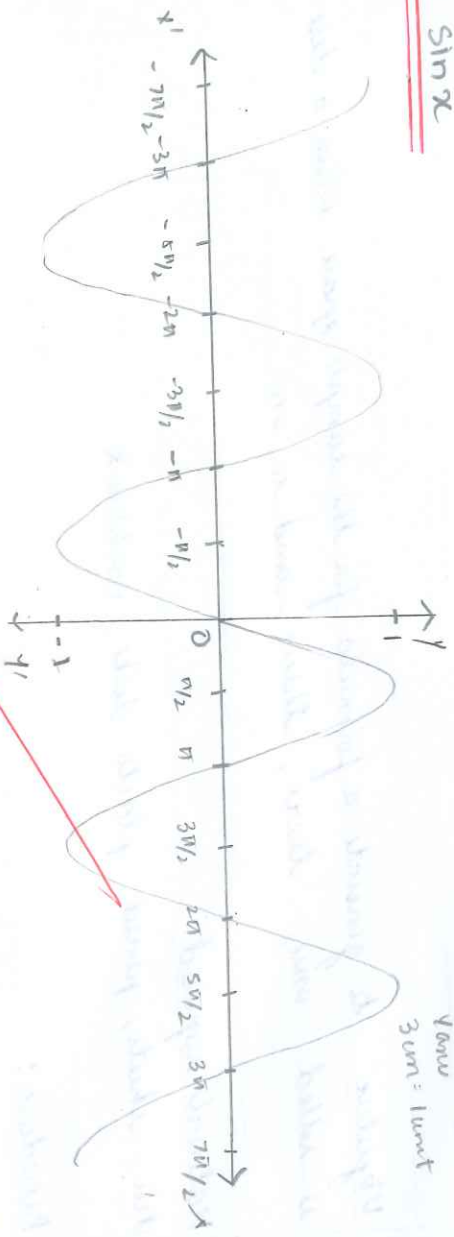
X axis:

Unit = $\pi/12$ radian

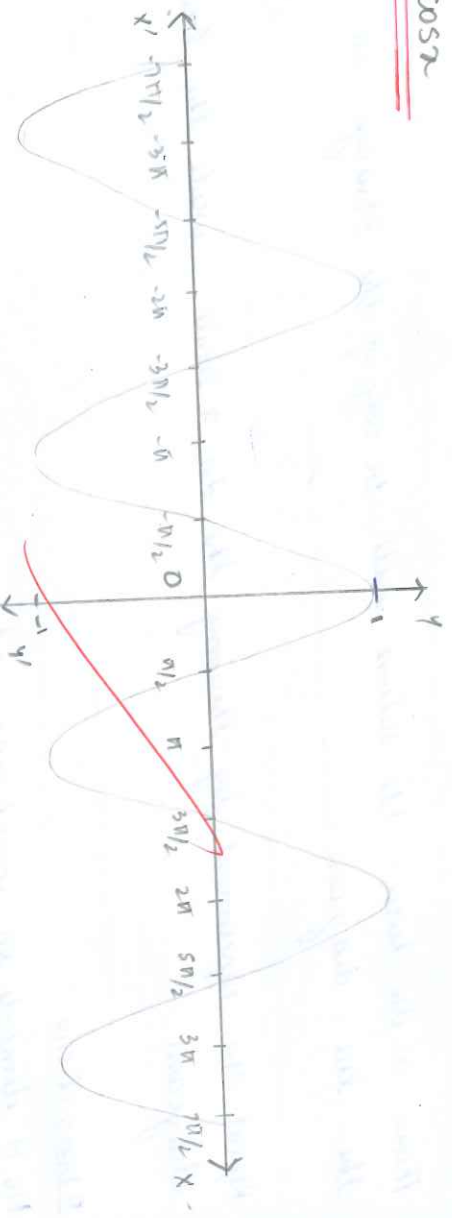
Y axis:

Unit = Unit

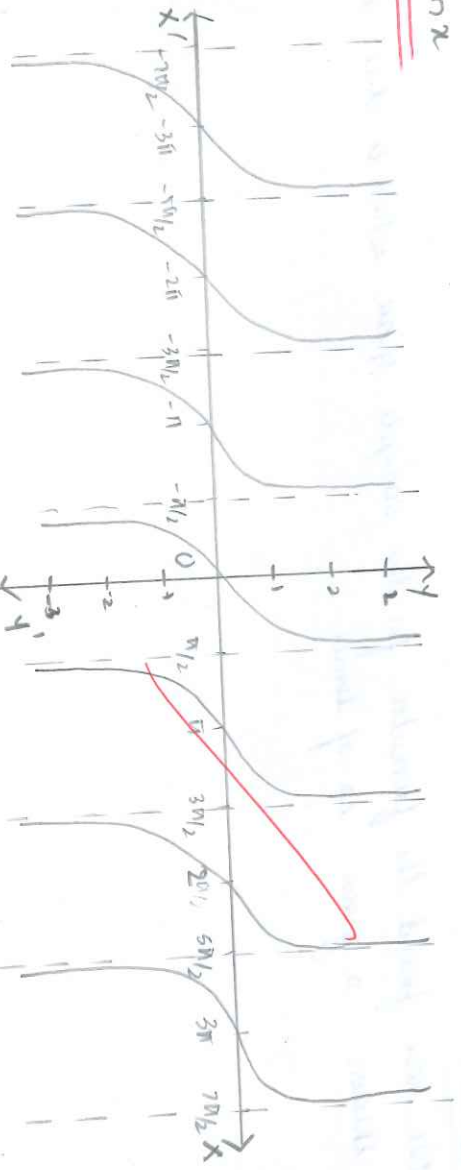
1) $\sin x$



2) $\cos x$



3) $\tan x$



GRAPHS OF TRIGONOMETRIC FUNCTIONSAim:

To draw graphs of trigonometric functions

Materials required:

paper, geometry box

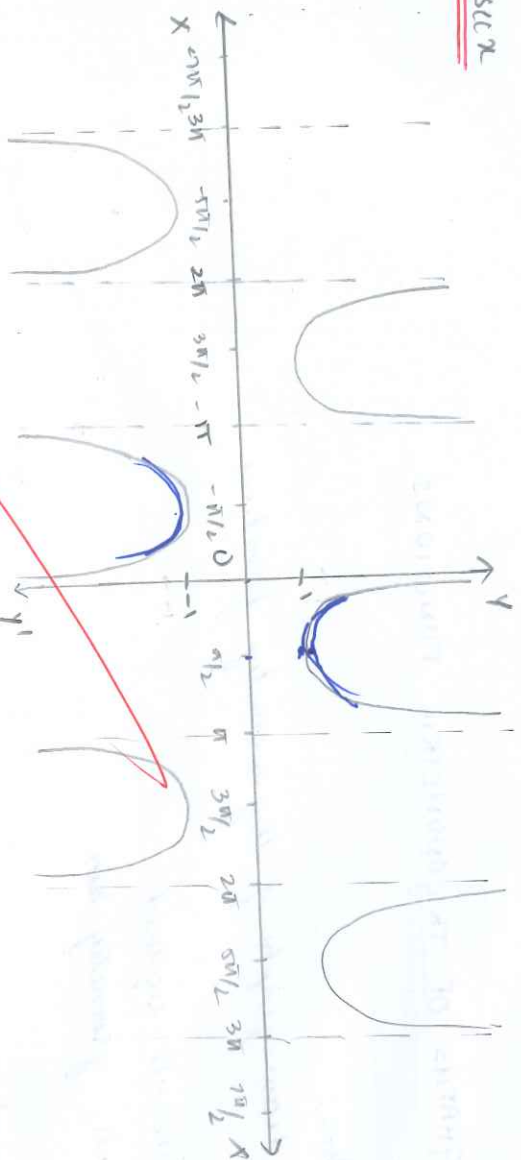
Procedure:

draw the graphs of sine, cosine, tangent, secant, cosecant, cotangent

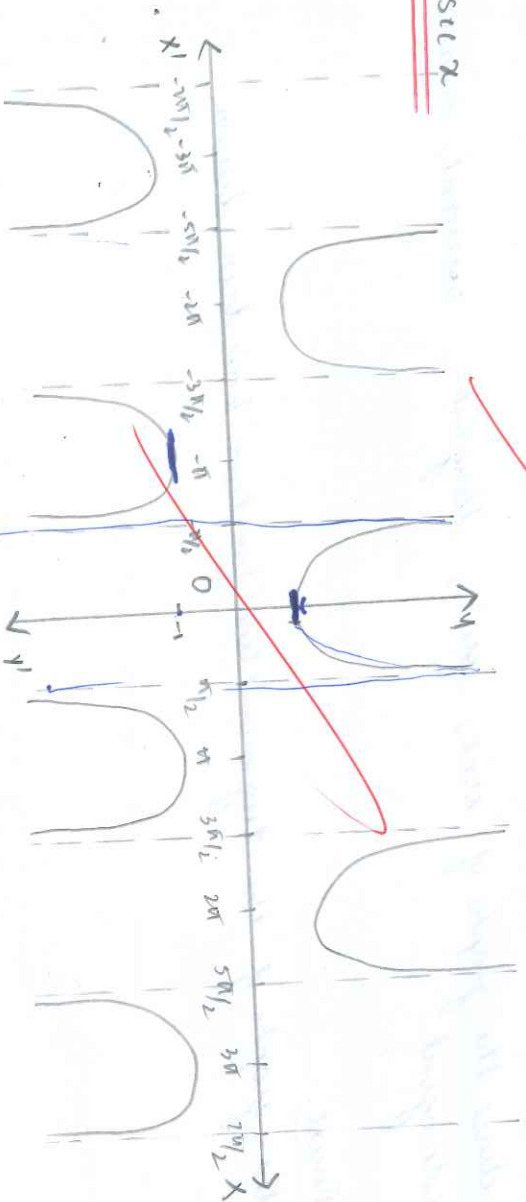
Result:

the graphs of the trigonometric functions have been drawn.

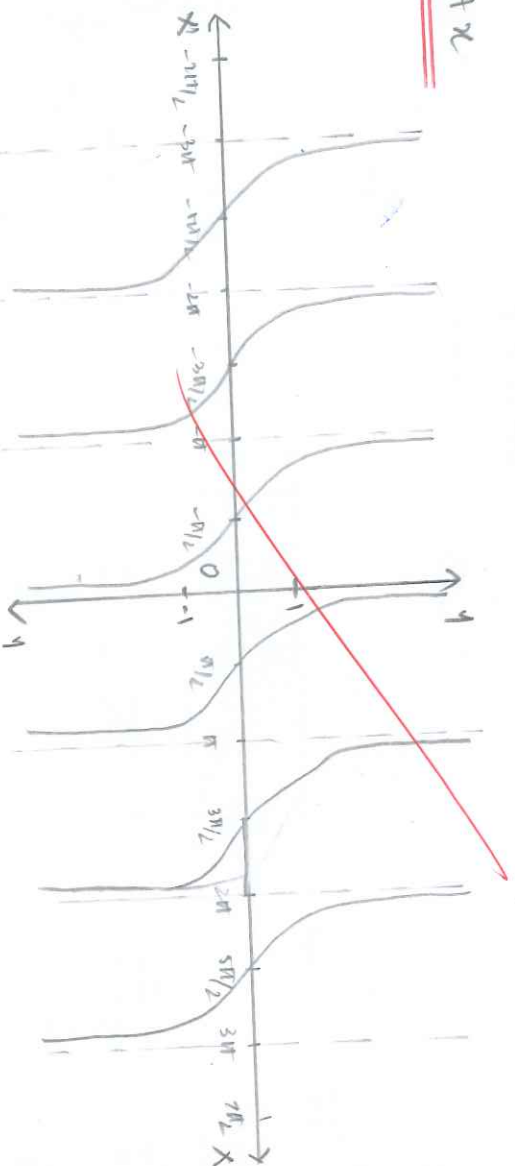
4) $\cos x$



5) $\sec x$



6) $\cot x$



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Domain and Range of Trigonometric functions :

	T - function	Domain	Range
1.	$\sin x$	R	$[-1, 1]$
2.	$\cos x$	R	$[-1, 1]$
3.	$\tan x$	$x \in R, x \neq (2n+1)\frac{\pi}{2}$	R
4.	$\cot x$	$x \in R : x \neq n\pi$	R
5.	$\csc x$	$x \in R : x \neq n\pi$	$R - (-1, 1)$
6.	$\sec x$	$x \in R : x \neq (2n+1)\frac{\pi}{2}$	$R - (-1, 1)$

D. M. M. M.